Programming Fundamentals

Week 6 — Building a CRUD Application

Programming Fundamentals Team

SETU

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Last Time

We combined concepts we already knew:

lists + dictionaries + loops + selection + functions

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We combined concepts we already knew:

lists + dictionaries + loops + selection + functions and introduced:

Create — Read — Update — Delete

Our Week 6 Project: Dog Day Care Manager





Imagine you work at a **Dog Day Care Centre**. Staff need a simple program to keep track of the dogs attending.

Each dog has an **ID, name, breed and age**.

Our program will let us **add, view, find, update and delete** dog records.

Our Week 6 Project: Dog Day Care Manager

The programming goal: practise CRUD using **lists of dictionaries, functions and a menu**.

Plan the Dog Records

Before we write the Python, describe the steps in plain English.

```
START with an empty collection of dogs
FOR EACH dog attending day care
    STORE its ID, name, breed and age together
ADD the dog record to the collection
```

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Now compare these steps with the Python on the next slide.

Our Dog Day Care Example

We will build a small **Dog Day Care Manager**.

```
dogs = [
    {"id": 1, "name": "Buddy", "breed": "Golden Retriever", "age": 4},
    {"id": 2, "name": "Luna", "breed": "Labrador", "age": 2}
]
```

Our Dog Day Care Example

We will build a small **Dog Day Care Manager**.

```
dogs = [
    {"id": 1, "name": "Buddy", "breed": "Golden Retriever", "age": 4},
    {"id": 2, "name": "Luna", "breed": "Labrador", "age": 2}
]
```

Today the data exists only while the program is running.

Plan: Display All Dogs

Before we write the Python, describe the steps in plain English.

```
RECEIVE the list of dogs
FOR EACH dog in the list
    DISPLAY its ID, name, breed and age
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Start with Read

Before changing data, make sure we can display it clearly.

```
def display_dogs(dogs):
    for dog in dogs:
        print(dog["id"], dog["name"], dog["breed"], dog["age"])
```



Start with Read

Before changing data, make sure we can display it clearly.

```
def display_dogs(dogs):
    for dog in dogs:
        print(dog["id"], dog["name"], dog["breed"], dog["age"])
```

Why pass `dogs` as a parameter?

Plan: Find a Dog by ID

Before we write the Python, describe the steps in plain English.

```
RECEIVE the list of dogs and an ID to search for
FOR EACH dog in the list
    IF the dog ID matches the requested ID
        RETURN that dog
RETURN None if no dog matches
```

Plan: Find a Dog by ID

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    IF the dog ID matches the requested ID
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RETURN None if no dog matches
```

Now compare these steps with the Python on the next slide.

Find One Record

Several CRUD operations need to find a dog first.

```
def find_dog(dogs, dog_id):
    for dog in dogs:
        if dog["id"] == dog_id:
            return dog

    return None
```

Find One Record

Several CRUD operations need to find a dog first.

```
def find_dog(dogs, dog_id):
    for dog in dogs:
        if dog["id"] == dog_id:
            return dog

    return None
```

Write the search once. Reuse it wherever it is needed.

Plan: Handle a Search Result

Before we write the Python, describe the steps in plain English.

```
SEARCH for the dog with ID 2
IF a matching dog was found
    DISPLAY the dog name
ELSE
    DISPLAY "Dog not found"
```

Plan: Handle a Search Result

Before we write the Python, describe the steps in plain English.

```
SEARCH for the dog with ID 2
IF a matching dog was found
    DISPLAY the dog name
ELSE
    DISPLAY "Dog not found"
```

Now compare these steps with the Python on the next slide.

Using None

```
dog = find_dog(dogs, 2)

if dog is not None:
    print(dog["name"])
else:
    print("Dog not found")
```

Using None

```
dog = find_dog(dogs, 2)

if dog is not None:
    print(dog["name"])
else:
    print("Dog not found")
```

`None` means that the function did not find a matching record.

Plan: Add a Dog

Before we write the Python, describe the steps in plain English.

```
ASK the user for the dog ID, name, breed and age
CREATE a dog record using those values
ADD the record to the list of dogs
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Create — Add a New Record

```
def add_dog(dogs):
    dog_id = int(input("ID: "))
    name = input("Name: ")
    breed = input("Breed: ")
    age = int(input("Age: "))

    dog = {
        "id": dog_id,
        "name": name,
        "breed": breed,
        "age": age
    }

    dogs.append(dog)
```

Plan: Avoid Duplicate IDs

Before we write the Python, describe the steps in plain English.

```
SEARCH the list for the new dog ID
IF a dog with that ID already exists
    DISPLAY "That ID already exists"
ELSE
    CREATE and ADD the new dog record
```

Plan: Avoid Duplicate IDs

Before we write the Python, describe the steps in plain English.

```
SEARCH the list for the new dog ID
IF a dog with that ID already exists
    DISPLAY "That ID already exists"
ELSE
    CREATE and ADD the new dog record
```

Now compare these steps with the Python on the next slide.

Create — What Could Go Wrong?

Before adding a dog, ask:

Should two dogs be allowed to have the same ID?

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```
existing = find_dog(dogs, dog_id)

if existing is not None:
    print("That ID already exists")
else:
    # create and append the new dog
```

Create — What Could Go Wrong?

Before adding a dog, ask:

Should two dogs be allowed to have the same ID?

```
existing = find_dog(dogs, dog_id)

if existing is not None:
    print("That ID already exists")
else:
    # create and append the new dog
```

We are combining **Create** with **Read/search**.

Plan: Update a Dog

Before we write the Python, describe the steps in plain English.

```
ASK for the ID of the dog to update
SEARCH for the dog using its ID
IF the dog was found
    ASK for the new age
    CHANGE the dog age
    DISPLAY "Dog updated"
ELSE
    DISPLAY "Dog not found"
```

Plan: Update a Dog

Before we write the Python, describe the steps in plain English.

```
ASK for the ID of the dog to update
SEARCH for the dog using its ID
IF the dog was found
    ASK for the new age
    CHANGE the dog age
    DISPLAY "Dog updated"
ELSE
    DISPLAY "Dog not found"
```

Now compare these steps with the Python on the next slide.

Update — Find, Then Change

```
def update_dog(dogs):
    dog_id = int(input("Dog ID: "))
    dog = find_dog(dogs, dog_id)

    if dog is not None:
        dog["age"] = int(input("New age: "))
        print("Dog updated")
    else:
        print("Dog not found")
```

Update — Find, Then Change

```
def update_dog(dogs):
    dog_id = int(input("Dog ID: "))
    dog = find_dog(dogs, dog_id)

    if dog is not None:
        dog["age"] = int(input("New age: "))
        print("Dog updated")
    else:
        print("Dog not found")
```

We don't create a new dictionary — we change the existing one.

Plan: Update Several Fields

Before we write the Python, describe the steps in plain English.

```
AFTER finding the dog
    ASK for the new name
    CHANGE the stored name
    ASK for the new breed
    CHANGE the stored breed
    ASK for the new age
    CHANGE the stored age
```

Plan: Update Several Fields

Before we write the Python, describe the steps in plain English.

```
AFTER finding the dog
    ASK for the new name
    CHANGE the stored name
    ASK for the new breed
    CHANGE the stored breed
    ASK for the new age
    CHANGE the stored age
```

Now compare these steps with the Python on the next slide.



Update More Than One Field?

We could also allow the user to update other values:

```
dog["name"] = input("New name: ")
dog["breed"] = input("New breed: ")
dog["age"] = int(input("New age: "))
```



Update More Than One Field?

We could also allow the user to update other values:

```
dog["name"] = input("New name: ")
dog["breed"] = input("New breed: ")
dog["age"] = int(input("New age: "))
```

Design question: should the user have to change **everything**?

There is often more than one reasonable solution.

Plan: Delete a Dog

Before we write the Python, describe the steps in plain English.

```
ASK for the ID of the dog to delete
SEARCH for the dog using its ID
IF the dog was found
    REMOVE the dog from the list
    DISPLAY "Dog deleted"
ELSE
    DISPLAY "Dog not found"
```

Plan: Delete a Dog

Before we write the Python, describe the steps in plain English.

```
ASK for the ID of the dog to delete
SEARCH for the dog using its ID
IF the dog was found
    REMOVE the dog from the list
    DISPLAY "Dog deleted"
ELSE
    DISPLAY "Dog not found"
```

Now compare these steps with the Python on the next slide.

Delete — Find, Then Remove

```
def delete_dog(dogs):
    dog_id = int(input("Dog ID: "))
    dog = find_dog(dogs, dog_id)

    if dog is not None:
        dogs.remove(dog)
        print("Dog deleted")
    else:
        print("Dog not found")
```

CRUD Functions

Our program now has clear operations:

```
add_dog()       → Create
display_dogs()  → Read
find_dog()      → Read/Search
update_dog()    → Update
delete_dog()    → Delete
```



CRUD Functions

Our program now has clear operations:

```
add_dog()       → Create
display_dogs()  → Read
find_dog()      → Read/Search
update_dog()    → Update
delete_dog()    → Delete
```

Each function has a clear responsibility.

Plan: Show a Menu

Before we write the Python, describe the steps in plain English.

```
DISPLAY the available options:
    Add dog, View dogs, Update dog, Delete dog, Exit
ASK the user to choose an option
STORE the choice
```

Plan: Show a Menu

Before we write the Python, describe the steps in plain English.

```
DISPLAY the available options:
    Add dog, View dogs, Update dog, Delete dog, Exit
ASK the user to choose an option
STORE the choice
```

Now compare these steps with the Python on the next slide.



How Does the User Choose?

We already know how to build this:

```
print("1. Add dog")
print("2. View dogs")
print("3. Update dog")
print("4. Delete dog")
print("0. Exit")

choice = input("Choose an option: ")
```



How Does the User Choose?

We already know how to build this:

```
print("1. Add dog")
print("2. View dogs")
print("3. Update dog")
print("4. Delete dog")
print("0. Exit")

choice = input("Choose an option: ")
```

What do we need if we want the menu to appear again?

Plan: Repeat the Menu

Before we write the Python, describe the steps in plain English.

```
SET choice to an empty value
WHILE choice is not "0"
    DISPLAY the menu options
    ASK the user for a choice
```

Plan: Repeat the Menu

Before we write the Python, describe the steps in plain English.

```
SET choice to an empty value
WHILE choice is not "0"
    DISPLAY the menu options
    ASK the user for a choice
```

Now compare these steps with the Python on the next slide.

A Menu Loop

```
choice = ""

while choice != "0":
    print("1. Add dog")
    print("2. View dogs")
    print("3. Update dog")
    print("4. Delete dog")
    print("0. Exit")

    choice = input("Choose an option: ")
```



A Menu Loop

```
choice = ""

while choice != "0":
    print("1. Add dog")
    print("2. View dogs")
    print("3. Update dog")
    print("4. Delete dog")
    print("0. Exit")

    choice = input("Choose an option: ")
```

This is why we learned `while` loops.

Plan: Act on the Menu Choice

Before we write the Python, describe the steps in plain English.

```
IF the choice is "1"
    ADD a dog
ELSE IF the choice is "2"
    DISPLAY all dogs
ELSE IF the choice is "3"
    UPDATE a dog
ELSE IF the choice is "4"
    DELETE a dog
ELSE IF the choice is not "0"
    DISPLAY "Invalid option"
```

Plan: Act on the Menu Choice

Before we write the Python, describe the steps in plain English.

```
IF the choice is "1"
    ADD a dog
ELSE IF the choice is "2"
    DISPLAY all dogs
ELSE IF the choice is "3"
    UPDATE a dog
ELSE IF the choice is "4"
    DELETE a dog
ELSE IF the choice is not "0"
    DISPLAY "Invalid option"
```

Now compare these steps with the Python on the next slide.

Connect the Menu to the Functions

```
if choice == "1":
    add_dog(dogs)
elif choice == "2":
    display_dogs(dogs)
elif choice == "3":
    update_dog(dogs)
elif choice == "4":
    delete_dog(dogs)
elif choice != "0":
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Connect the Menu to the Functions

```
if choice == "1":
    add_dog(dogs)
elif choice == "2":
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    delete_dog(dogs)
elif choice != "0":
    print("Invalid option")
```

Selection decides which operation to perform.

Look at What We Have Combined

Our small CRUD application uses:

Concept	Job in the program
Variables	Store individual values
Selection	Choose actions and handle decisions
Functions	Organise operations
Iteration	Repeat menus and process records
Lists	Store the collection
Dictionaries	Represent each record



Important Limitation

Add a dog, then stop the program and run it again.

What happens?



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Add a dog, then stop the program and run it again.

What happens? **The new dog is gone.**

Important Limitation

Add a dog, then stop the program and run it again.

What happens? **The new dog is gone.** Our list only exists in memory while Python is running.

This gives us our next problem to solve: persistence using files / JSON.

Mini Activity — Trace a CRUD Operation

Choose one operation: **Create, Read, Update or Delete**.

Write its steps in plain English before looking at the Python.

Example for Update:

```
1. Ask for the ID
2. Search for the record
3. If found, ask for the new value
4. Change the record
5. Otherwise display an error
```

Mini Activity — Trace a CRUD Operation

Choose one operation: **Create, Read, Update or Delete**.

Write its steps in plain English before looking at the Python.

Example for Update:

```
1. Ask for the ID
2. Search for the record
3. If found, ask for the new value
4. Change the record
5. Otherwise display an error
```

Then identify where selection, iteration and functions are needed.

Week 6 — Key Takeaways

- CRUD is a pattern for managing collections of data
- CRUD is not a new Python feature
- It combines concepts we already know
- Searching is an important reusable operation
- Functions keep each operation manageable
- Our data currently disappears when the program ends

Week 6 — Key Takeaways

- CRUD is a pattern for managing collections of data
- CRUD is not a new Python feature
- It combines concepts we already know
- Searching is an important reusable operation
- Functions keep each operation manageable
- Our data currently disappears when the program ends

We now have the foundations for a much more realistic data-driven program.